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REVIEW EXPERIMENTAL/SPECIAL TOPICS

Anatomy of Forehead, Glabellar, Nasal and Orbital Muscles, and Their Correlation with Distinctive Patterns of Skin Lines on the Upper Third of the Face: Reviewing Concepts



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Abstract
Background The purpose of this study is to establish a relationship between the skin lines on the upper third of the face in cadavers, which represent the muscle activity in life and the skin lines achieved by voluntary contraction of the forehead, glabellar, and orbital muscles in patients.
Methods Anatomical dissection of fresh cadavers was performed in 20 fresh cadavers, 11 females and 9 males, with ages ranging from 53 to 77 years. Subcutaneous dissection identified the muscle shape and the continuity of the fibers of the eyebrow elevator and depress muscles. Subgaleal dissection identified the cutaneous insertions of the muscles. They were correlated with skin lines on the upper third of the face of the cadavers that represent the muscle activity in life. Voluntary contraction was performed by 20 voluntary patients, 13 females and 7 males, with ages ranging from 35 to 62 years. Distinct patterns of skin lines on the forehead, glabellar and orbital areas, and eyebrow displacement were identified.
Results The frontalis exhibited four anatomical shapes with four different patterns of horizontal parallel lines on the forehead skin. The corrugator supercilii showed three shapes of muscles creating six patterns of vertical glabellar lines, three symmetrical and three asymmetrical. The orbicularis oculi and procerus had single patterns. The skin lines exhibited in voluntary contraction of the upper third of the face in patients showed the same patterns of the skin lines achieved in cadavers.
Conclusions Skin lines in cadavers, which are the expression of the muscle activity in life, were similar to those achieved in the voluntary contraction of patients, allowing us to assert that the muscle patterns of patients were similar to those identified in cadavers.
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Keywords Anatomical dissection · Voluntary contraction · Frontalis · Corrugator supercilii · Procerus · Orbicularis oculi

Introduction

The muscles of the facial expression on the upper third of the face are separated into two antagonistic groups. The eyebrow elevator muscle composed of the medial, intermediate and lateral fibers of the frontalis, and the eyebrow depressor muscles composed by procerus, corrugator supercilii, and orbicularis oculi [1]. They have common properties such as cutaneous insertions and continuity between fibers at the glabella and orbital rim. Cutaneous insertions are responsible for forehead and glabellar skin lines, and orbital crow's feet. In turn, the continuity of the fibers and the antagonism between the elevator and depressor groups guide the shape and location of the eyebrow [2]. Distinctive frontalis muscle contraction patterns

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achieved through voluntary contraction were correlated to different anatomical shapes of the frontalis [3]. Frontalis activity was also evaluated relating electromyography measurements and eyebrow displacement. It was found that the left and right frontalis generated the same amount of muscle activity [4]. The influence of the corrugator supercilii has been relevant in forehead and glabellar expression, and in eyebrow shape and location. Variabilities in shape and location of the corrugator supercilii have been described either as a short narrow muscle located at the medial end of the eyebrow, or as a long straight muscle along the medial half of the eyebrow [5]. However, there is some controversy regarding the muscles working at the medial end of the eyebrow and glabella. The depressor supercilii, a distinct muscle from the corrugator supercilii and orbicularis oculi, has been reported as a depressor of the eyebrow medial end and responsible for glabellar frown lines [6]. Specific topographic analysis reported that the dimensions of the corrugator supercilii can be easily delineated using fixed bony landmarks such as the orbital rim at its medial end [7]. A recent review of the anatomy of the corrugator supercilii stated that glabellar frown lines are formed by repeated contractions of this muscle [8]. The involvement of the orbicularis oculi in the lowering of the eyebrow lateral end was widely emphasized in lateral orbicularis oculi muscle plasty [9]. The relationship of the procerus with the horizontal lines on the roof of the nose has been highlighted through electromyographical tests [10].

The purpose of this study is to establish a relationship between the skin lines on the upper third of the face in cadavers, which represent muscle activity in life, with the skin lines achieved by voluntary contraction of the forehead, glabellar, and orbital muscles in patients. This relationship will allow us to correlate the anatomical aspects of the forehead, glabellar, and orbital muscles in patients with those achieved in cadavers.

Methods

To better understand the anatomy and expressiveness of the muscles on the upper third of the face, two distinct ways of assessment, related one to another, were employed. Anatomical dissection of fresh cadavers was performed to identify the shape and attachments of the forehead, glabellar, and orbital muscles that were correlated with the skin lines on the upper third of the face. Another appraisal was the voluntary contraction of the forehead, glabellar, and orbital muscles in patients that identified extension, direction, and location of the skin lines achieved, as well as eyebrow displacement. The Pearson product-moment correlation coefficient was used to compare skin lines in cadavers with those achieved in voluntary contraction in patients. It was done in order to establish a correlation between the muscle patterns achieved in cadavers with the muscles of the patients. The Mann–Whitney test investigated any statistically significant difference among the samples.

Anatomical Dissection Findings

Dissection was performed in twenty fresh cadavers, 11 females and 9 males, with ages ranging from 53 to 77 years and mean age of 58.0 (±7.41) years. Forty frontalis, procerus, corrugator supercilii, and orbicularis oculi muscles were isolated from the surrounding tissues. Photos from the forehead, glabella, eyelid, and eyebrows of the specimens were taken to relate the skin lines on the upper third of the face with the shape and location of the dissected muscles. Dissection was carried out first in the subgaleal level showing the continuity between the forehead, glabellar, and orbital muscle fibers. Next, the frontalis was released from the procerus, corrugator supercilii, and pars orbitalis of the orbicularis oculi, exhibiting the cutaneous insertions of each muscle (Fig. 1). After that, a subcutaneous dissection was performed exposing the shape, dimension, and insertions of each muscle. It also exposed the continuity between the frontalis lateral fibers with those of the orbicularis oculi, frontalis intermediate fibers with those of the corrugator supercilii, and frontalis anteromedial fibers with those of the procerus (Fig. 2).

Voluntary Muscle Contraction and Lines of Facial Expression

Twenty voluntary patients, 13 females and 7 males, with ages ranging from 35 to 62 years and mean age of 42.90 (±8.54) years, were chosen to evaluate the facial expression on the upper third of the face. Exclusion criteria

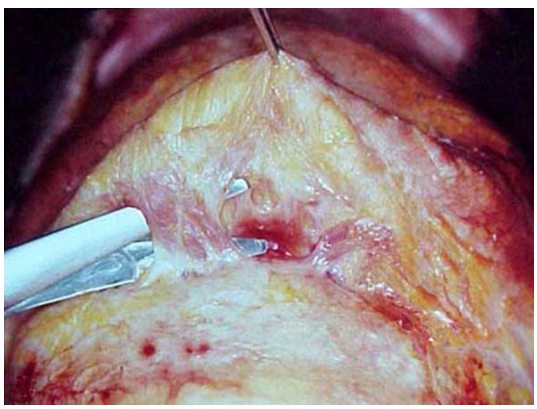


Fig. 1 Subgaleal dissection exposed cutaneous insertions of the depressor muscles

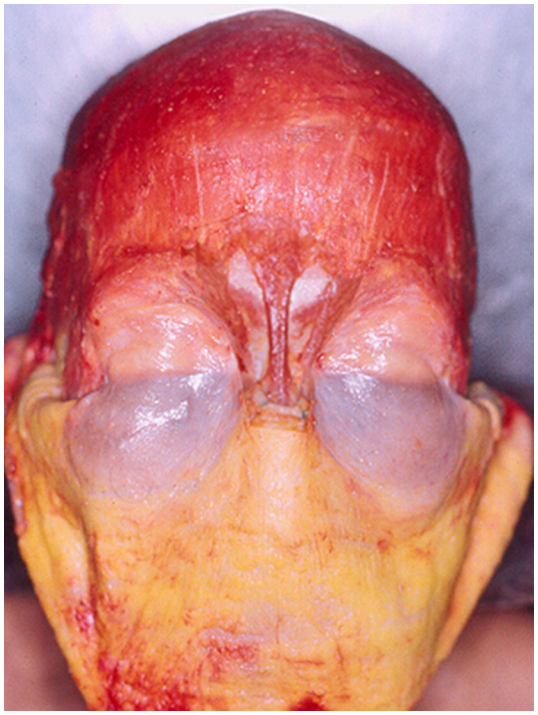


Fig. 2 Subcutaneous dissection identified the continuity between the elevator and depressor muscles of the upper third of the face

included prior surgical or cosmetic treatments on the upper third of the face. All patients signed an informed consent form, agreeing with the procedure performed according to the Helsinki Declaration. The patients were recorded and photographed during three specific movements: raising the forehead; frowning the glabella; and by forcing a smile. The movements were performed separately and repeated five times each. Patterns of facial expression were identified and classified according to the eyebrow movements and arrangement of the forehead, glabellar, and eyelid lines.

Results

The lines on the forehead, glabella, and lateral eyelid in the cadavers were marks left on the skin by the muscle activity in life. The same patterns of lines on the forehead, glabella, and lateral eyelids in cadavers were encountered in voluntary contraction of the patients. Such correlation led us to understand that each type of muscle produces its own pattern of muscle contraction. Table 1 summarizes the types of muscle patterns and outcomes in cadavers. Table 2 summarizes distinct patterns of voluntary contraction and outcomes in patients. The Pearson linear correlation coefficient from −1 to 1 showed a significant correlation, of 1, between cadavers and patients for the four patterns of forehead lines. The Mann–Whitney test showed no statistically significant differences with p value = 0.056 between cadavers and patients for the six patterns of glabellar lines.

Anatomical Dissection Findings

Frontalis two bellies, one on each side of the forehead, arose from the aponeurotic galea, at the level of the coronal suture, inserting into the deep layer of the forehead skin. Four distinct patterns for frontalis were identified and each one of them established a different pattern of horizontal parallel lines on forehead skin. *Type I or full form* was the most frequent pattern, found in 9/20 (45 %) of specimens; the bellies are joined in the middle line located on the entire forehead (Fig. 3), being responsible for straight lines along the forehead. *Type II or V-shaped form* was found in 6/20 (30 %) of specimens; the bellies are medially separated by a V-shaped projection of the aponeurotic galea (Fig. 4), creating gull wing-shaped lines on the forehead (Fig. 5). *Type III or central form* was found in 2/20 (10 %) of specimens; the bellies are joined in the middle line, over the medial half of both orbital rims (Fig. 6), originating a column of short lines on the middle forehead. *Type IV or lateral form* was found in 3/20 (15 %) of specimens; the bellies are located on the lateral half of the orbital rim separated medially by a vertical rectangular projection of the aponeurotic galea (Fig. 7), addressing a column of short lines on each lateral end of the forehead (Fig. 8).

Procerus two small pyramidal muscles from the lower part of the nasal bone to the skin between the eyebrows, exhibiting only one pattern, identified as *Single Type* (Fig. 9). They addressed horizontal lines on the roof of the nose.

Orbicularis oculi arose from the frontal process of the maxilla and nasal process of the frontal bone running under the corrugator supercilii (Fig. 10). Two patterns were identified: *Single Type I*—the thicker peripheral fibers were inserted in the deep layer of the eyebrow lateral end, being responsible for its lowering— and *Single Type II*—the thinner central fibers were inserted in the deep layer of the palpebral skin, creating eyelid crow's feet.

Corrugator supercilii arose from the frontal process of the maxilla to the deep layer of the medial half of the eyebrow. Six muscle patterns were identified in cadaver specimens: three symmetrical and three asymmetrical. Three symmetrical patterns with the same shape, dimension, and insertion on both sides of the glabella were classified as follows: *Type I or fan-shaped form* was the most frequent symmetrical pattern, found in 6/20 (30 %) of specimens, with insertion along the entire medial half of the eyebrow (Fig. 11), creating a hockey stick-shaped line on the medial end of the eyebrows (Fig. 5). *Type II or rectangular form* was found in 4/20 (20 %) of specimens,

Table 1 Distribution of the cadavers according to age, sex, frontalis, corrugator supercilii, procerus, and orbicularis oculi

	Age	Sex	Frontalis		Corrugator supercilii		Procerus	Orbicularis
			Muscle type	Cutaneous lines	Muscle type	Cutaneous lines		
Cadaver 1	53	M	Full	Horizontal straight lines	Fan-shaped	Fold-shaped hockey stick	Single type	Single type
Cadaver 2	75	F	Full	Horizontal straight lines	Type VI	Type II + Type III	Single type	Single type
Cadaver 3	63	F	V-shaped	Line-shaped gull wing	Type VI	Type II + Type III	Single type	Single type
Cadaver 4	57	F	Lateral form	Lateral straight lines	Fan-shaped	Fold-shaped hockey stick	Single type	Single type
Cadaver 5	67	M	Full	Horizontal straight lines	Fan-shaped	Fold-shaped hockey stick	Single type	Single type
Cadaver 6	72	M	Central form	Central straight lines	Triangular form	Parallel straight creases	Single type	Single type
Cadaver 7	77	M	Full	Horizontal straight lines	Fan-shaped	Fold-shaped hockey stick	Single type	Single type
Cadaver 8	61	F	V-shaped	Line-shaped gull wing	Type VI	Type I + Type II	Single type	Single type
Cadaver 9	76	F	V-shaped	Line-shaped gull wing	Type VI	Type I + Type III	Single type	Single type
Cadaver 10	56	M	Full	Horizontal straight lines	Fan-shaped	Fold-shaped hockey stick	Single type	Single type
Cadaver 11	75	F	Full	Horizontal straight lines	Type VI	Type II + Type III	Single type	Single type
Cadaver 12	64	F	Central form	Central straight lines	Type VI	Type I + Type III	Single type	Single type
Cadaver 13	64	F	Full	Horizontal straight lines	Fan-shaped	Fold-shaped hockey stick	Single type	Single type
Cadaver 14	66	M	Lateral form	Lateral straight lines	Type VI	Type II + Type III	Single type	Single type
Cadaver 15	57	M	Lateral form	Lateral straight lines	Type VI	Type I + Type II	Single type	Single type
Cadaver 16	59	F	V-shaped	Line-shaped gull wing	Type VI	Type II + Type III	Single type	Single type
Cadaver 17	63	F	V-shaped	Line-shaped gull wing	Triangular form	Parallel straight creases	Single type	Single type
Cadaver 18	55	M	V-shaped	Line-shaped gull wing	Triangular form	Parallel straight creases	Single type	Single type
Cadaver 19	62	F	Full	Horizontal straight lines	Type VI	Type I + Type II	Single type	Single type
Cadaver 20	61	M	Full	Horizontal straight lines	Narrow ribbon	Single-straight crease	Single type	Single type

Table 2 Distribution of the patients according to sex, age, and voluntary contraction of frontalis, corrugator supercilii, procerus, and orbicularis

	Sex	Age	Frontalis	Corrugador supercilii	Procerus	Orbicularis
Patient 1	F	41	Horizontal straight lines	Fold-shaped hockey stick	Single type	Single type
Patient 2	M	47	Horizontal straight lines	Fold-shaped hockey stick	Single type	Single type
Patient 3	F	36	Horizontal straight lines	Parallel straight creases	Single type	Single type
Patient 4	M	54	Central straight lines	Type I + Type II	Single type	Single type
Patient 5	F	36	Horizontal straight lines	Type I + Type II	Single type	Single type
Patient 6	M	62	Horizontal straight lines	Single-straight crease	Single type	Single type
Patient 7	F	37	Line-shaped gull wing	Type II + Type III	Single type	Single type
Patient 8	M	37	Horizontal straight lines	Fold-shaped hockey stick	Single type	Single type
Patient 9	F	45	Lateral straight lines	Fold-shaped hockey stick	Single type	Single type
Patient 10	M	39	Lateral straight lines	Parallel straight creases	Single type	Single type
Patient 11	F	35	Horizontal straight lines	Single-straight crease	Single type	Single type
Patient 12	F	41	Lateral straight lines	Single-straight crease	Single type	Single type
Patient 13	M	39	Line-shaped gull wing	Type I + Type III	Single type	Single type
Patient 14	F	44	Central straight lines	Type I + Type II	Single type	Single type
Patient 15	M	42	Horizontal straight lines	Type II + Type III	Single type	Single type
Patient 16	F	35	Horizontal straight lines	Parallel straight creases	Single type	Single type
Patient 17	F	53	Line-shaped gull wing	Parallel straight creases	Single type	Single type
Patient 18	F	36	Line-shaped gull wing	Fold-shaped hockey stick	Single type	Single type
Patient 19	F	37	Line-shaped gull wing	Type I + Type II	Single type	Single type
Patient 20	F	62	Line-shaped gull wing	Fold-shaped hockey stick	Single type	Single type



Fig. 3 Frontalis muscle is joined in the middle line involving the entire forehead characterizing Type I or full form



Fig. 4 Type II or V-shaped frontalis muscle is identified by an inverted triangular projection of the aponeurotic galea between both bellies of the frontalis

with insertion in the intermediate third of the eyebrow (Fig. 12), creating parallel straight lines on the glabella. *Type III or narrow ribbon* was found in 1/20 (5 %) of



Fig. 5 Gull wing-shaped lines on the forehead of the cadaver, caused by Type II or V-shaped frontalis muscle. Symmetrical hockey stick-shaped lines on the medial end of the eyebrows, created by Type I or fan-shaped corrugator supercilii muscle

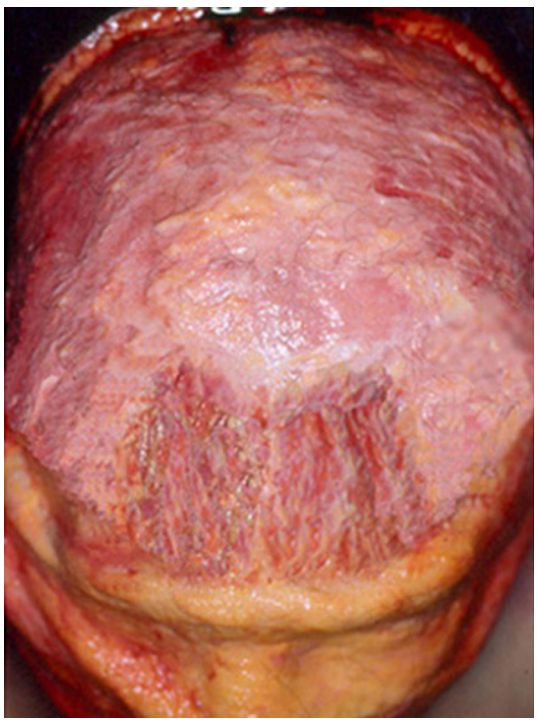


Fig. 6 Type III or Central form of the frontalis muscle is a variation of Type I, not involving the lateral portion of the forehead

specimens, with insertion in the medial end of the eyebrow (Fig. 13), creating a single vertical line on the glabella. Three asymmetrical patterns having different shapes, sizes, and insertions on each side of the glabella were classified as follows: *Type IV or asymmetry I* was the most frequent asymmetrical pattern, found in 5/20 (25 %) of specimens, with a fan-shaped form on one side and a rectangular form on the other, creating a hockey stick shape and parallel



Fig. 7 Type IV or Lateral form of the frontalis muscle is a variation of Type II with the muscles separated at the middle line by a rectangular projection of the aponeurotic galea



Fig. 8 Column of short lines on each side of the forehead at its lateral portion, caused by Type IV or lateral form of the frontalis muscle. A hockey stick-shape on the *right* side of the glabella and parallel straight lines on the *left*, created by corrugator supercilii asymmetrical Type IV or asymmetry I with a fan-shaped form on one side and a rectangular form on the other

straight lines, respectively (Fig. 8). *Type V or asymmetry II* was found in 2/20 (10 %) of specimens having a fan-shaped form on one side and a narrow ribbon on the other, creating a hockey stick-shaped line and single-straight line, respectively. *Type VI or asymmetry III* was found in 3/20 (15 %) of specimens having a rectangular form on one side

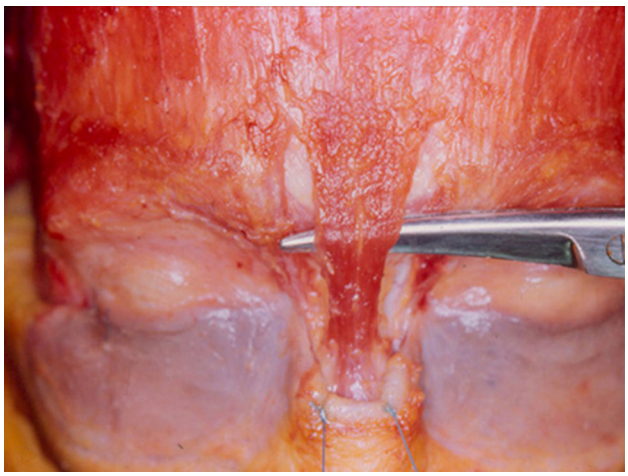


Fig. 9 Shape and bone insertions of both procerus muscles and continuity with the medial fibers of the frontalis

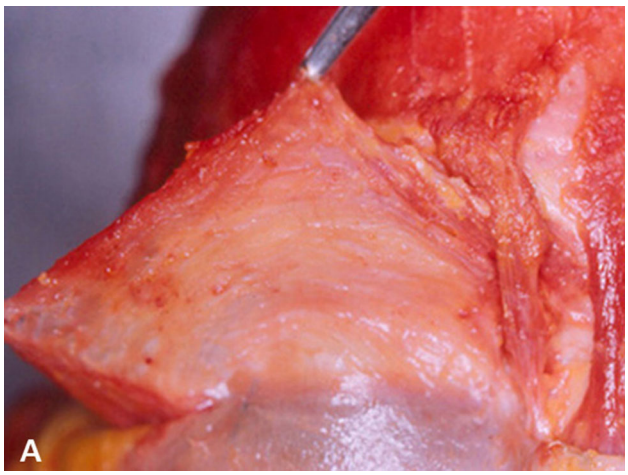


Fig. 10 Orbicularis oculi running underneath the corrugator supercilii. The orbital portion of the orbicular oculi is thicker at the entire orbital rim and thin on the eyelid

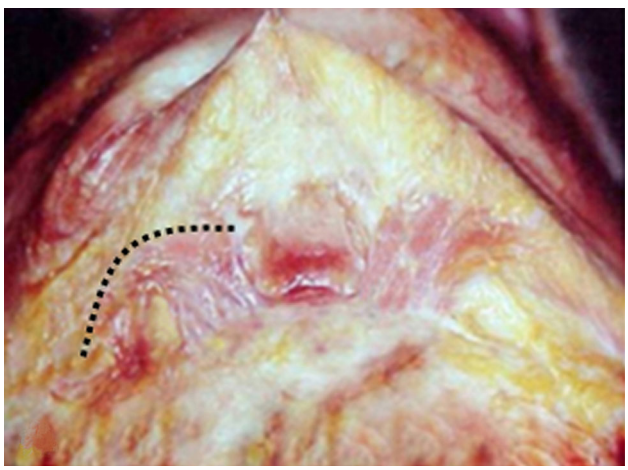


Fig. 11 Type I or fan-shaped corrugator supercilii showing the muscle insertion in the deep layer of the medial half of the eyebrow

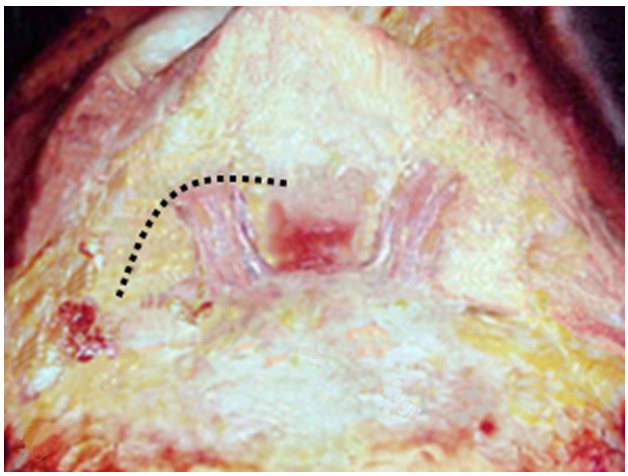


Fig. 12 Type II or rectangular corrugator supercilii with insertion in the intermediate portion of the medial half of the eyebrow

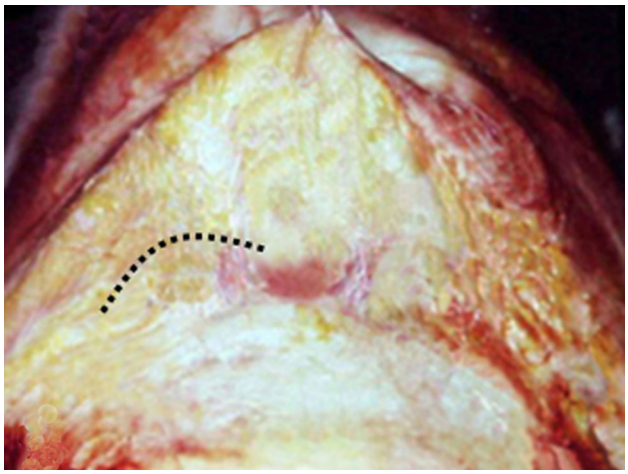


Fig. 13 Type III or narrow ribbon corrugator supercilii is a muscle strip inserted in the medial end of the eyebrow

and a narrow ribbon on the contralateral side, creating several parallel straight lines on the glabella.

Voluntary Muscle Contraction and Lines of Facial Expression

Frontalis showed four patterns of parallel horizontal lines on the forehead: *Type I or full straight lines* showed parallel horizontal lines across the forehead in 9/20 patients (45 %) (Fig. 14). *Type II or gull wing-shaped lines* were found on the entire forehead in 6/20 patients (30 %) (Fig. 15). *Type III or short central straight lines* exhibited a middle forehead column in 2/20 patients (10 %) (Fig. 16). *Type IV or short lateral straight lines* showed two columns on each lateral end of the forehead in 3/20 patients (15 %) (Fig. 17).

Procerus lowered the skin between the eyebrows in all patients, causing horizontal lines on the roof of the nose classified as *Single Type*.

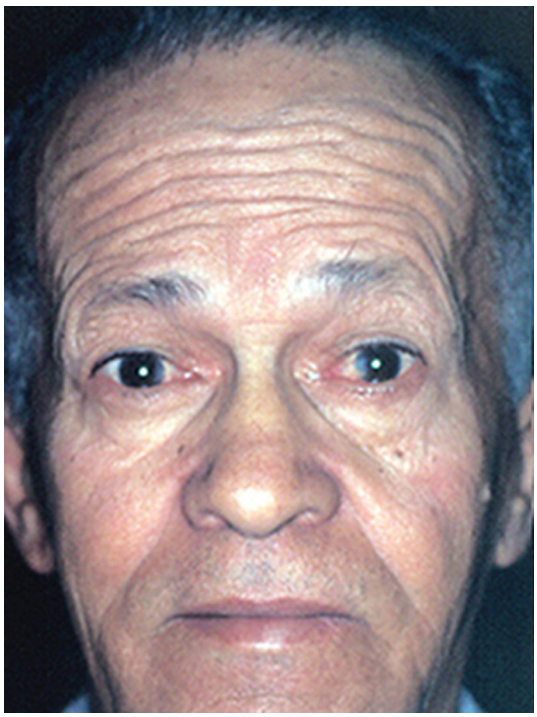


Fig. 14 Type I or full straight lines of voluntary contraction of frontalis

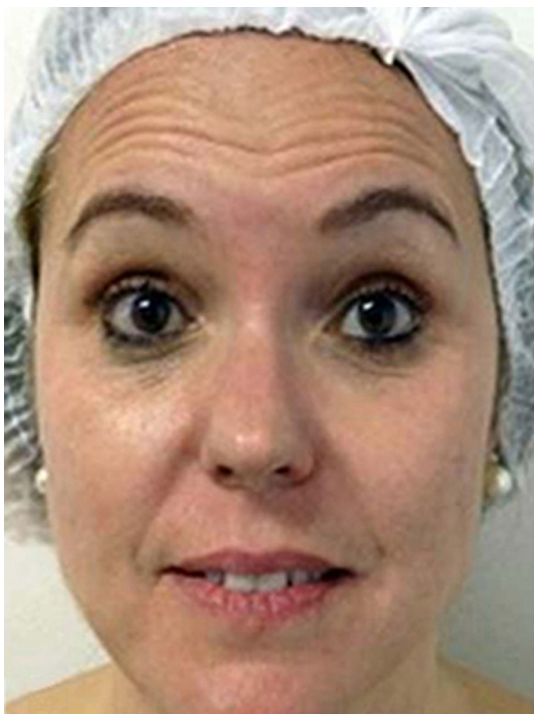


Fig. 15 Type II or line-shaped gull wing of voluntary contraction of frontalis

Orbicularis oculi two patterns of muscle contraction were identified in all patients: *Single Type I* lowered the lateral end of the eyebrow and *Single Type II* produced lateral eyelid crow's feet.



Fig. 16 Type III or short central column of straight lines of voluntary contraction of frontalis

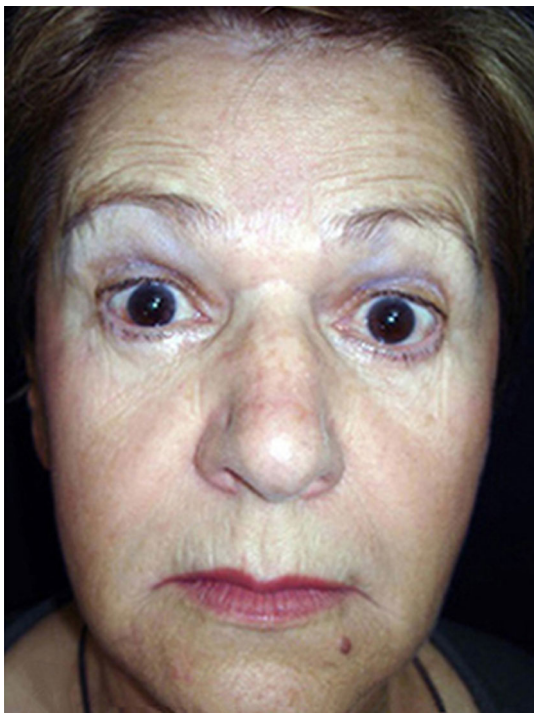


Fig. 17 Type IV or short lateral columns of straight lines of voluntary contraction of frontalis

Corrugator supercilii six patterns of vertical glabellar lines, three symmetrical and three asymmetrical patterns, were identified. Symmetrical patterns showed the same vertical lines on both sides of the glabella and were classified as follows: *Type I or hockey stick-shaped line* caused by medial approach and lowering of the eyebrows in 6/20 (30 %) patients (Fig. 18); *Type II or parallel straight lines* produced by midline approach of the eyebrows in 4/20 (20 %) patients (Fig. 19); and *Type III or single-straight line* created by lowering both medial ends of the eyebrows in 3/20 (15 %) patients (Fig. 20). Asymmetrical patterns reproduced vertical lines of different symmetrical patterns on each side of the glabella and were classified as follows: *Type IV* exhibited a hockey stick-shaped line and parallel straight lines in 4/20 (20 %); *Type V* showed a hockey stick-shaped line and single-straight line in 1/20 (5 %); and *Type VI* presented several parallel straight lines in 2/20 (10 %) patients.

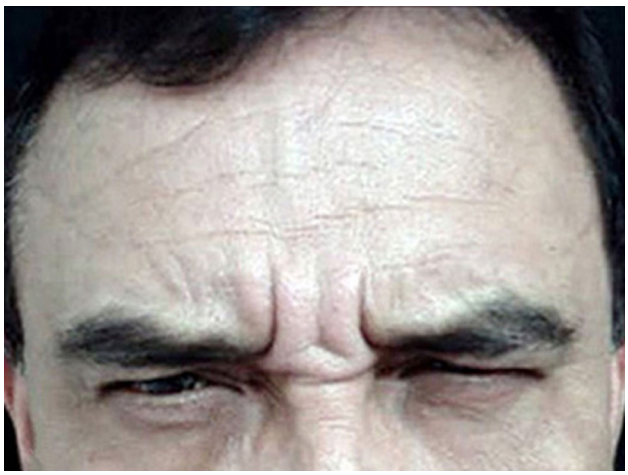


Fig. 18 Fold-shaped hockey stick glabellar lines resulting from voluntary contraction of the corrugator supercilii identified as Type I pattern



Fig. 19 Parallel straight creases expressed by voluntary contraction showed corrugator supercilii Type II pattern



Fig. 20 A single-straight crease between the eyebrows shown during voluntary contraction of the corrugator supercilii represents the Type III pattern of the muscle

Discussion

Several misconceptions were related to the anatomy and activity of the forehead, glabellar, and orbital muscles. An anatomical dissection associated with lines of facial expression was performed to better understand the anatomy and activity of these muscles. It was done to optimize the results of surgical and nonsurgical treatments of the aging face. To establish a relationship between skin lines achieved by voluntary contraction on the upper third of the face in patients with muscle patterns, a correlation was made with the skin lines on the upper third of the face in cadavers and corresponding muscle patterns found in anatomical dissection. This correlation was only possible because the skin lines in cadavers represent the muscle activity in life. To decrease the error rate, the cadavers were selected with ages from 53 to 77 years so as to exhibit well-defined skin lines. For patients, the selected age was between 35 and 62 years because skin lines are expressed by the force of the voluntary contraction of the muscle. Recently, an anatomical study reported asymmetry of the frontalis bellies in 6/29 cadavers [11]. Unlike, symmetry of the frontalis bellies was achieved in 20/20 specimens in this series of cadavers. Symmetry of the frontalis bellies was also widely evidenced in all frontalis patterns of voluntary contraction in this series of patients. Moreover, symmetrical patterns of forehead lines on both sides of the forehead in cadavers were identical to those achieved in forehead voluntary contraction in patients, showing that there is no asymmetry between the frontalis bellies. Furthermore, frontalis muscle activity measured with surface electromyography showed that both bellies of the frontalis generated the same amount of muscle activity [12]. Guerrerosantos et al. [13] reported that distinct eyebrow movements, and the glabellar frown lines are carried out by the corrugator supercilii. Controversies regarding eyebrow movements and glabellar lines involved certain homologies such as the corrugator supercilii, depressor supercilii, and medial head of the orbital portion of the orbicularis oculi muscles. These muscles were reported to be responsible for depression of the medial head of the eyebrow and to contribute to the formation of oblique glabellar skin lines [14]. Regardless of the name, eyebrow movements and glabellar lines are related to the dermal insertions of the muscle fibers. Lockhart et al. [15] described the corrugator supercilii as the medial head of the orbital portion of the orbicularis oculi responsible for lowering of the medial end of eyebrow. Unlike, in this series of cadavers, the medial head of the orbicularis oculi was observed underneath the corrugator supercilii without cutaneous insertion at the medial end of the eyebrow and therefore unable to lower the eyebrow. The ribbon-shaped depressor supercilii has been described as a distinct muscle from the corrugator supercilii with a definite, individual effect on the movement of the eyebrow, and skin of the glabella [16]. In this series of patients, a narrow ribbon-shaped muscle from the frontal process of the maxilla to the deep layer of the medial end of the eyebrow, was responsible for the lowering of the eyebrow medial end and identified as corrugator supercilii type III. Also, cadaver dissection showed only an origin for the corrugator supercilii with three distinct dermal insertions demonstrating that there were three distinct shapes for the same muscle. Another point of controversy regarding the depressor supercilii was the anatomical topography of the muscle. According to Sobotta and Becher [17], the depressor supercilii muscle is a small, narrow, pyramidal muscle, at the medial end of the eyebrow, beneath the frontalis belly. As the depressor of supercilii has the frontalis interposed between itself and the eyebrow, it is not inserted in the deep layer of the skin at the medial end of the eyebrow and therefore cannot depress the eyebrow. The glabellar lines in cadavers, expressions of the muscle activity in life, were similar to those of voluntary contractions in patients, suggesting that distinct types of corrugator supercilii caused distinct patterns either for eyebrow movements or for glabellar lines. They were symmetrical for corrugator supercilii Types I, II, and III because the muscles were equal on both sides of the glabella and asymmetrical in Types IV, V, and VI because the muscles had distinct types on each side of the glabella. Recent anatomical findings showed that the corrugator supercilii ran over the medial head of the orbital portion of the orbicularis oculi to be inserted into the deep layer of the skin along the medial half of the eyebrow [18].

Conclusion

We concluded that cutaneous lines in cadavers, expressions of the muscle activity in life, were similar to those achieved in the voluntary contractions of patients, suggesting that the muscle patterns identified in cadavers were similar to those of the patients.

Compliance with Ethical Standards

Conflict of Interest The authors have no financial interest or conflict of interest to declare in relation to the content of this article.

Ethical Approval "To the cadavers of selfless dedication" and "Declare that all procedures involving human participants were in accordance with the ethical standards of the Institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards."

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